Lab 14 - Ironing - Glass-Smooth Top Surfaces

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 03: Process Improvement for AM TSC objective: A5 - apply ironing to improve top-surface quality and assess its cost

Goal

Enable Cura's ironing feature, which passes the hot nozzle over the top surface with a tiny extrusion, and compare the top finish with and without it.

What you'll build

Two cube prints - ironed and standard - demonstrating the top-surface improvement.

Tools: Ultimaker Cura, cube10.stl, z-block.stl, FDM printer

Steps

1. Open cube10.stl and slice with the standard 0.2 mm profile; print cube 1 as the control.

```
Slice > print control cube
```

2. Enable ironing.

```
Top/Bottom > Enable Ironing ✔
```

3. Inspect the ironing pass in Preview - a dense zigzag over the top face at ~10% flow.

```
Preview > top layer > ironing pass
```

4. Slice and note the added time (ironing is slow but only on top faces).

```
Slice > record extra minutes
```

5. Print cube 2 with ironing enabled.

```
Save to Disk > print
```

6. Compare the two top faces under raking light: the ironed face should be near-flat with faint tool lines.

```
Visual + touch comparison
```

7. Repeat the Preview comparison with z-block.stl to inspect how ironing travels around raised detail; print it only if time permits.

```
Open z-block.stl > Enable Ironing > Preview top layers
```

Evidence to capture

- Control-versus-ironed time/material record.
- Raking-light photos comparing the flat and detailed top surfaces.

Test it

The ironed cube is visibly smoother than the control, z-block.stl shows ironing constrained to eligible top faces, and you can justify the added time for lids, logos and presentation surfaces.

Files in this folder

- cube10.stl - Core sample model used by the lab procedure.
- z-block.stl - Detailed top-surface benchmark for evaluating ironing around raised geometry.